Physics-Guided ML for PV Performance Under Partial Shading

Research Project Roadmap • Sprint Plan • Experimental Blueprint • Paper Strategy

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Project objective: Investigate where a simplified photovoltaic physics model deviates from a higher-fidelity reference under irradiance, temperature and partial-shading conditions, and determine whether a residual machine-learning model can correct those errors while retaining the structure and interpretability of the physics model.

Important research framing: Do not treat this as simply an MLP power-prediction project. The central scientific investigation is the *physics-model error*, its dependence on operating regimes, and whether residual learning can correct it—including under out-of-distribution (OOD) conditions.

1. Research Questions

RQ1 — Physics behavior: In which irradiance, temperature and partial-shading regimes does the analytical model deviate from a higher-fidelity reference?

RQ2 — Pure ML: Can a purely data-driven model predict PV maximum power accurately across the studied regimes?

RQ3 — Hybrid correction: Can ML learn only the residual error of the physics model and improve prediction while preserving the physics model as the base?

RQ4 — OOD generalization: How does the physics model, pure ML model and hybrid model behave when tested on regimes excluded from training?

2. Core Research Hypothesis

> **Hypothesis:** The simplified physics model will show systematic, regime-dependent discrepancies under complex partial shading and temperature combinations. A residual ML model may learn these discrepancies more effectively than directly replacing the physics model, but its benefit and robustness must be evaluated inside and outside the training distribution.

3. Proposed Modeling Framework

Physics baseline: $P_{physics}$ is generated using an analytical single-diode PV model.

High-fidelity reference: $P_{reference}$ is generated from a more detailed array-level simulation that represents module-level irradiance/shading and electrical mismatch behavior.

Residual: $\Delta P = P_{reference} - P_{physics}$

Residual ML: $\Delta P_{ML} = f_{\theta}(X)$

Hybrid prediction: $P_{hybrid} = P_{physics} + \Delta P_{ML}$

Pure ML baseline: $P_{ML} = g_{\theta}(X)$

> **Critical terminology:** In the final paper, refer to the single-diode model as the *physics baseline*, not automatically as ground truth. Use the higher-fidelity simulation as the reference.

4. System and Dataset Definition

Initial controlled simulation domain:

Parameter	Initial range / design
PV array	12 panels/modules

Irradiance	100–1200 W/m²
Temperature	0–60 °C
Shaded panels	Random 1–4 panels
Shading severity	20–80%
Shading representation	Panel-level shading mask / fractional irradiance
Primary target	Pmax

Recommended feature representation: include irradiance G, temperature T, number of shaded modules, shading severity and the individual panel-level shading mask S1…S12. Preserve the actual shading configuration rather than storing only the number of shaded panels.

5. Three Required Model Families

Model A — Physics only: analytical baseline. Establishes how accurate the physics approximation is.

Model B — Pure ML: direct prediction of Pmax from operating/shading features. Establishes the black-box baseline.

Model C — Hybrid residual ML: physics prediction + learned residual. This is the proposed method.

Minimum comparison table for the final paper:

Model	IID test	Temperature OOD	Shading OOD	Combined OOD
Physics	MAE/RMSE	MAE/RMSE	MAE/RMSE	MAE/RMSE
Pure ML	MAE/RMSE	MAE/RMSE	MAE/RMSE	MAE/RMSE
Hybrid residual	MAE/RMSE	MAE/RMSE	MAE/RMSE	MAE/RMSE

6. Research-Grade Train/Test Design

Avoid relying only on a random train_test_split. Random splitting can place almost identical operating points in both training and testing sets and hide extrapolation failures.

IID/interpolation test: random samples within the training domain.

Temperature holdout: reserve low/high temperature bands for testing.

Shading holdout: train on moderate severity and reserve more severe shading for evaluation.

Combined OOD: reserve combinations of extreme irradiance, temperature and shading.

> **Principle:** OOD regions must be defined before final testing, not selected after seeing which cases produce favorable results.

7. Experimental Matrix

The final experiment should answer not only “which model has lower error?” but also “under which physical regimes does each model succeed or fail?”

Experiment	Purpose
Physics baseline	Quantify discrepancy of simplified analytical model.
Pure ML	Measure black-box predictive performance.
Residual ML	Test whether learning physics error is easier/more robust.
IID evaluation	Measure interpolation performance.
Temperature OOD	Test temperature extrapolation.
Shading OOD	Test severity/configuration extrapolation.

Combined OOD	Stress-test generalization.
Ablation	Identify which inputs/features contribute to residual correction.
Explainability	Understand which physical variables drive model error/correction.

8. Metrics and Analysis

Use MAE, RMSE, NRMSE and R^2. Treat MAPE cautiously because it becomes unstable when true power is close to zero.

Primary analysis: absolute and relative error distributions across physical regimes.

Error maps: visualize $|P_{reference} - P_{physics}|$ over irradiance × temperature and other relevant dimensions.

Residual plots: ΔP versus irradiance, temperature, shading severity and number of shaded panels.

Example curves: show representative I–V and P–V curves for unshaded, moderate-shading and severe-shading cases.

9. Ablation and Explainability

Recommended ablations:

1. Physics only.
2. Pure ML.
3. Full residual hybrid.
4. Residual model without temperature.
5. Residual model without panel-level shading configuration.
6. Residual model using only G, T, number of shaded panels and average severity.
7. Full panel-level shading mask.

Potential interpretation tools: permutation importance, SHAP, partial-dependence analysis and feature ablation. Any ranking must be reported from actual experiments rather than assumed beforehand.

10. Software / Compute Stack

Core: Python, NumPy, Pandas, SciPy, Matplotlib, Seaborn, Jupyter.

PV modeling: pvlib-python for analytical/single-diode modeling; PVMismatch or another validated array-level approach for partial-shading/mismatch reference modeling.

ML: Scikit-learn for conventional baselines and evaluation; PyTorch for the residual MLP if appropriate.

Interpretability: SHAP or comparable feature-analysis methods.

11. Scrum / Semester Roadmap

Sprint 0 — Research preparation

Literature review, research questions, system definition, repository, dataset schema and experimental protocol.

Sprint 1 — Physics + dataset foundation

Implement physics model; establish higher-fidelity reference; generate 5k–10k development samples; verify physically sensible behavior; produce initial error maps; prepare blueprint presentation.

Sprint 2 — ML + hybrid development

Implement pure ML and residual ML; tune models; establish evaluation pipeline; compare physics vs pure ML vs hybrid.

Sprint 3 — OOD + ablation

Run held-out regime experiments, extreme OOD tests, ablations and interpretability analyses.

Midsem milestone

Presentation + demo + preliminary paper: introduction, related work, methodology, experimental setup and preliminary results.

Final phase

Scale experiments if justified, validate externally where compatible, perform statistical/robustness analysis, finalize paper and presentation.

12. Sprint 1 — Exact Deliverables

Sprint 1 should demonstrate that the project is scientifically defined and technically feasible—not that the final ML model is already complete.

Deliverable	What to show
Research question	RQ1–RQ4 and central hypothesis.
Literature	15–25 relevant papers organized into physics, ML, hybrid and residual-learning themes.
PV architecture	12-panel topology, module parameters, shading representation and assumptions.
Physics baseline	Working single-diode model producing I–V/P–V curves and Pmax.
Reference model	Initial high-fidelity/array-level simulation capable of producing reference Pmax.
Dataset v1	5k–10k development simulations with documented schema.
Residual analysis	$\Delta P = P_{reference} - P_{physics}$ and first error visualizations.
Experimental plan	IID, regime-held-out and OOD split definitions.
Repository	Organized code, configuration, notebooks, literature and results.
Presentation	10–12 slides with architecture, workflow and initial evidence.

13. Sprint 1 Presentation — Recommended 12 Slides

Slide 1: Title + team

Slide 2: Problem motivation

Slide 3: Research gap

Slide 4: Research questions + hypothesis

Slide 5: Proposed physics $\rightarrow$ residual ML framework

Slide 6: 12-panel PV system and assumptions

Slide 7: Dataset generation space

Slide 8: Three-model comparison: Physics / Pure ML / Hybrid

Slide 9: IID and OOD experimental design

Slide 10: Evaluation metrics + planned visualizations

Slide 11: Expected scientific contribution and limitations

Slide 12: Sprint roadmap + responsibilities

14. First 7–10 Days — Action Plan

Days 1–2: Literature search and paper matrix. Target 15–25 strong papers.

Days 2–3: Freeze the 12-panel electrical topology, module parameters, shading representation and simulation assumptions.

Days 3–5: Implement and validate the analytical physics model.

Days 5–7: Implement partial shading and reference simulation; verify different shading masks produce sensible behavior.

Days 7–8: Generate first 5k–10k samples and save a versioned dataset.

Days 8–9: Compute residuals and make error plots/heatmaps plus representative I–V/P–V curves.

Day 10: Freeze Sprint 1 blueprint and presentation.

15. Recommended GitHub Structure

PV-Hybrid-ML/
■■■ README.md
■■■ literature/
■■■ configs/
■■■ data/
■■■ notebooks/
■■■ src/
■ ■■■ physics/
■ ■■■ simulation/
■ ■■■ models/
■ ■■■ evaluation/
■ ■■■ visualization/
■■■ experiments/
■■■ results/
■■■ paper/

16. Research Quality Rules

Rule 1: Do not optimize only for a high R^2. Explain where and why models fail.

Rule 2: Do not use only random train/test splitting. Reserve physically meaningful regimes.

Rule 3: Do not claim the hybrid model is superior before experiments establish it.

Rule 4: Document every simulation assumption and parameter source.

Rule 5: Keep code, datasets, configurations and experiment seeds/versioning reproducible.

Rule 6: Treat OOD performance as a primary research result, not an afterthought.

Rule 7: Use AI transparently as required by the course; cite AI assistance where used and ensure every team member understands submitted code for the viva.

17. Potential Scientific Contribution

> If supported by the experiments, the paper could contribute a regime-based analysis of simplified PV physics-model error and an evaluation of residual learning as a correction mechanism under partial shading and temperature variation, including its limitations under OOD conditions.

Important: The final contribution must be based on what the experiments actually discover. Do not predetermine that ML will outperform physics or that the hybrid model will generalize better.

18. Paper Structure

1. Abstract
2. Introduction
3. Related Work
4. PV Physics Model
5. Dataset and Simulation Methodology
6. Proposed Residual Hybrid Model
7. Experimental Design and OOD Protocol
8. Results
9. Ablation and Explainability
10. Discussion and Limitations
11. Conclusion
12. References

19. Initial Reference/Tool Targets

• pvlib-python — analytical PV and single-diode modeling.

• PVMismatch — PV mismatch / array-level partial-shading modeling.

• NIST PV array performance data — potential external measured-data validation source.

• NREL NSRDB — potential irradiance/weather source for later realism checks.

• Literature on physics-guided ML, hybrid PV modeling and residual learning — used to establish the research gap.

20. One-Line Project Strategy

Do not build an ML model that replaces PV physics; build an experiment that discovers where the physics model fails, then test whether ML can learn and correct that failure without throwing away the physics.